

KYC CHATBOT - ARCHITECTURE SUMMARY

Project

- Name: KYC Document Chatbot
- Type: Production-level Hybrid RAG chatbot for KYC documents

Core Stack

- Backend: FastAPI + LangChain
- Frontend: React + Tailwind + Vite
- LLM: OpenAI GPT-4o
- Embeddings: text-embedding-3-small
- Vector DB: ChromaDB
- Retrieval: Hybrid (Dense + Sparse) with Re-ranking

High-Level Flow

1. User sends query from frontend.
2. FastAPI backend receives query.
3. Hybrid retrieval pipeline runs:
 - Dense retrieval via ChromaDB (semantic search)
 - Sparse retrieval via BM25 (keyword search)
4. Reciprocal Rank Fusion (RRF) merges dense and sparse results.
5. FlashRank (ms-marco-MiniLM-L-12-v2) re-ranks merged candidates.
6. Top relevant chunks are sent as context to GPT-4o.
7. Final response + sources returned to UI.

Retrieval Pipeline Details

- Dense Search (ChromaDB)
 - Uses OpenAI embeddings for semantic similarity.
 - Captures intent and meaning beyond exact keywords.
- Sparse Search (BM25)
 - Uses lexical keyword matching and TF-IDF style scoring.
 - Strong for exact terms, entities, and regulation keywords.
- Fusion (RRF)
 - Combines both result lists using reciprocal-rank scoring.
 - Reduces bias of single retriever and improves robustness.
- Re-ranking (FlashRank)
 - Model: ms-marco-MiniLM-L-12-v2
 - Lightweight CPU-friendly relevance re-ranker.
 - Improves precision before final LLM response generation.

Document Ingestion Flow

1. Upload document (TXT/PDF/DOCX/MD)
2. Validate and extract text
3. Chunk text
4. Generate embeddings
5. Store in ChromaDB
6. Build/refresh BM25 index for sparse retrieval

Deployment Architecture

- Frontend deployed on Vercel:
<https://kyc-bot.vercel.app/>
- Backend API deployed on Render:
<https://kyc-chatbot-api.onrender.com>

API Endpoints (Key)

- POST /api/v1/chat
- POST /api/v1/documents/upload
- GET /api/v1/documents
- GET /health

Why This Architecture

- Hybrid retrieval improves recall and robustness.
- Re-ranking improves final context precision.
- Guardrails + source-based responses reduce hallucination risk.
- Suitable for enterprise KYC document Q&A scenarios.

Repository

<https://github.com/Charul00/KYC-BOT>